Relational Database Schema & Normalization Note

- Data Dictionary & Detailed Table Definitions

3.1. Entity: students (Student Profile)

Represents registered undergraduate students authorized to request lab and workshop allocations.

Attribute	Data Type	Constraint / Key	Null?	Description
index_no	TEXT	PRIMARY KEY	No	Unique student identification / registration number.
name	TEXT	NOT NULL	No	Full legal name of the registered student.
department	TEXT	NOT NULL	No	Academic department the student belongs to.
email	TEXT	NOT NULL, UNIQUE	No	Official institutional email address for notifications.

3.2. Entity: workshops (Workshops & Laboratories)

Represents physical laboratory or workshop facilities eligible for reservation.

Attribute	Data Type	Constraint / Key	Null?	Description
category	TEXT	PRIMARY KEY	No	Unique workshop or lab category name (e.g., Electronics Lab).
department	TEXT	NOT NULL	No	Department responsible for managing and supervising the lab.

3.3. Entity: records (Booking / Allocation Master Record)

Represents the primary master transaction header created when a student submits an allocation request.

Attribute	Data Type	Constraint / Key	Null?	Description
id	INTEGER	PRIMARY KEY AUTO	No	Surrogate unique auto-increment identifier for the booking.
index_no	TEXT	FOREIGN KEY	No	Student who placed reservation (REFERENCES students.index_no).
category	TEXT	FOREIGN KEY	No	Reserved lab category (REFERENCES workshops.category).
supervisor_name	TEXT	NOT NULL	No	Name of supervising lecturer / project consultant.
project_name	TEXT	NOT NULL	No	Official title of the coursework or research project.

Attribute	Data Type	Constraint / Key	Null?	Description
record_date	TEXT	NOT NULL	No	Scheduled calendar date for facility use (YYYY-MM-DD).
description	TEXT	DEFAULT NULL	Yes	Optional detailed justification or purpose for access.
passcode	TEXT	NOT NULL	No	Unique security passcode / allocation token for verification.
created_at	TEXT	NOT NULL	No	System timestamp of request submission (ISO 8601 string).
status	TEXT	DEFAULT 'Pending'	No	Workflow status ('Pending', 'Approve', 'Reject').
admin_remarks	TEXT	DEFAULT NULL	Yes	Optional administrator feedback, approval notes, or rejection reason.

3.4. Entity: booking_slots (Time Slot Child / Weak Entity)

Represents individual reservation intervals associated with a master booking. Decomposing slots into this child entity eliminates multi-valued attributes in records.

Attribute	Data Type	Constraint / Key	Null?	Description
record_id	INTEGER	PK, FK (CASCADE)	No	Master booking reference (REFERENCES records.id ON DELETE CASCADE).
time_slot	TEXT	PRIMARY KEY	No	Designated reservation time window (e.g., '08:30 - 10:30').

Referential Integrity & Relationship Constraints

- Students → Records (1:M): One student can have many bookings. Each booking belongs to one student. The foreign key records.index_no ensures that the student exists.
- Workshops → Records (1:M): One workshop can have many bookings. Each booking belongs to one workshop. The foreign key records.category ensures that the workshop exists.
- Records → Booking_Slots (1:M): One booking can have one or more time slots. The booking_slots table stores these slots separately. ON DELETE CASCADE automatically deletes the related slots when a booking is deleted.

Normalization Analysis (1NF, 2NF, 3NF)

Normalization is the process of organizing a database to reduce data duplication and avoid data problems. The proposed database satisfies 3NF.

First Normal Form (1NF)

A table is in 1NF when:

- Each field contains only one value.
- There are no repeating groups.
- Each record has a unique primary key.

The booking_slots table stores each time slot separately instead of storing multiple time slots in one field. Therefore, the database satisfies 1NF.

Second Normal Form (2NF)

A table is in 2NF when:

- It is already in 1NF.
- Every non-key attribute depends on the whole primary key.

The students, workshops, and records tables have single-column primary keys. The booking_slots table has a composite key (record_id, time_slot) and contains no non-key attributes. Therefore, the database satisfies 2NF.

Third Normal Form (3NF)

A table is in 3NF when:

- It is already in 2NF.
- There are no transitive dependencies.
- Non-key attributes depend only on the primary key.

Student details are stored in the students table, while workshop details are stored in the workshops table. These details are not unnecessarily repeated in the records table. Therefore, there are no transitive dependencies and the database satisfies 3NF.